



SysAdmin Master Runbook

Windows / Network / Identity / Database / Cloud / Operations

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Prepared for: Infrastructure & DBA Operations

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1. Master Architecture

This runbook covers the full GRT infrastructure stack end to end: Windows platform administration, network services, identity, applications, virtualization, database, cloud, and daily operations. Each domain below links to its detailed processes, checklists, and reference tables in later sections.

1.1 Domain Map

Domain	Components
Windows	CMD, PowerShell, Services, Processes, Event Logs, Registry, Storage, Performance
Network	IP, DNS, DHCP, Ports, Routing, Firewall, SMB
Identity	Active Directory, Users, Groups, GPO, Permissions
Application	IIS, Certificates, Logs, Runtime
Virtualization	Hyper-V, VMs
Database	MSSQL, SQL Agent, Backup, Restore, Blocking, Performance
Cloud	Azure VM, NSG, Storage, Monitoring, Backup
Operations	Monitoring, Incident, RCA, Change, Backup, DR

1.2 Layered Stack (top to bottom escalation path)

When troubleshooting, walk the stack in this order — it mirrors how a request actually travels through the infrastructure:

USER



APPLICATION



SERVER



PROCESS



SERVICE



PORT



FIREWALL



NETWORK



DNS



DATABASE



STORAGE



OPERATING SYSTEM



HARDWARE / CLOUD

2. Server Provisioning Process

Standard flow: Server → Script → CSV. Automated by New-ServerProvisioning.ps1 (see Automation Toolkit appendix).

2.1 The 14 Steps

#	Step	Automated?
1	Confirm hostname	Yes
2	Confirm IP configuration	Yes
3	Confirm DNS	Yes
4	Confirm time synchronization	Yes
5	Apply approved patches	Yes (requires PSWindowsUpdate/WSUS)
6	Install required roles/features	Yes
7	Configure firewall	Yes
8	Configure monitoring	Partial - agent check only
9	Configure backup	Partial - feature check only
10	Create required service accounts	Manual - AD approval gate
11	Configure storage	Yes (reporting)
12	Install application/database	Manual - role-specific
13	Test connectivity	Yes
14	Document configuration	Yes - appends to inventory CSV

2.2 Server Inventory Schema (CSV)

Column headers: ServerName, Environment, Role, IP, OS, Role, Owner, Application, Database, Location, Backup, Monitoring

ServerName	Environment	Role	IP	Owner
WEB01	Production	IIS	10.0.1.10	ApplicationTeam
DB01	Production	SQL	10.0.1.20	DBA

3. Identity & Access Onboarding

Automated by New-UserOnboarding.ps1. Manager approval is a hard gate — the script will not create an AD account without it.

HR Request



Manager Approval



Create AD Account



Assign Required Groups



Mailbox/Application Access



Computer Assignment



VPN/Remote Access



MFA



Test Login



Document

4. Application Deployment Flow

Application files



Configuration



Dependencies



Service/IIS



Port



Database



Authentication



Application test

4.1 Web Request Path (browser to database)

Browser



DNS



Server



Port 80/443



Firewall



IIS



Website



Application Pool



Application



Database

5. Monitoring & Alerting

5.1 Data Sources to Escalation

Windows Metrics / SQL Metrics / Application Logs / Security Events / Network Metrics



Monitoring



Alerts



SysAdmin



Incident

5.2 What Gets Monitored

- CPU, RAM, Disk
- Services, Ports, Certificates
- Backups, SQL jobs, Database state
- Event logs, Application errors

5.3 Threshold Bands

Metric	Normal	Warning	Critical
CPU	< 75%	75-90%	> 90%
Disk free	> 20%	10-20%	< 10%
Certificate expiry	> 30 days	< 30 days	Expired
Backup	Successful	Warning (slow/partial)	Failed

6. Incident Severity Matrix

Severity	Definition
SEV1	Production completely unavailable
SEV2	Major business outage
SEV3	Major functionality unavailable
SEV4	Multiple users affected
SEV5	Limited functionality
SEV6	Individual/team impact
SEV7	Information / Request
SEV8	Minor issue

7. Change Management Process

Request



Risk Assessment



Approval



Backup



Implementation



Validation



Monitoring



Close Change

8. Security Baseline Checklist

Apply to every server, application, and automated access device before go-live and revisit during quarterly access reviews.

- Least privilege
- Strong authentication
- MFA where supported
- Patch management
- Firewall enabled
- Endpoint protection
- Secure remote administration
- Audit logs
- Backup protection
- Restore testing
- Certificate monitoring
- Account lifecycle management
- Regular access review

8.1 Data/Server Flow for Agent-Based Monitoring

Server



Agent



Monitoring



Alert

9. Daily Operations Checklist

9.1 Morning (automated by Daily-HealthCheck.ps1)

- Check monitoring dashboard
- Check critical alerts
- Check server availability
- Check CPU/RAM
- Check disk
- Check critical services
- Check backups
- Check SQL jobs
- Check certificates
- Review overnight incidents

9.2 During the Day

- Process tickets
- User access
- Application issues
- Network issues
- Server changes
- Database issues
- Security alerts
- Document changes

9.3 End of Day

- Verify critical services
- Verify backups
- Review unresolved incidents
- Update tickets
- Document changes
- Handover

10. Troubleshooting Decision Trees

10.1 Disk Full (automated by Disk-Space-Triage.ps1)

Disk full



Which volume?



Which directory?



Which file type?



Which application owns it?



Is it log/data/backup/temp?



Retention policy?



Archive/expand/cleanup



Validate

10.2 Scheduled Task Failure

Task exists?



Enabled?



Trigger correct?



Run-as account valid?



Password/credential valid?



Program path valid?



Working directory valid?



Permissions?



Log/output?

10.3 Network Connectivity

NIC



Driver



Cable/Wi-Fi



Switch



Duplex/speed



Network congestion

10.4 Data Retention / Deletion

Identify



Confirm retention policy



Backup/archive if required



Approval



Delete



Verify

11. Reference Tables

11.1 Key Windows Security Event IDs

Event ID	Meaning
4624	Successful logon
4625	Failed logon
4634	Logoff
4648	Explicit credential logon
4672	Special privileges assigned
4688	Process creation
4720	User account created
4725	User account disabled
4726	User account deleted
4740	Account locked out

11.2 File Type Reference by Platform

Platform	Extensions
Windows	.exe, .msi, .msix, .ps1, .bat, .cmd, .dll, .sys
macOS	.dmg, .pkg, .app
Linux	.deb, .rpm, .ApplImage, .sh
Android	.apk, .aab
Archives	.zip, .7z, .rar, .tar, .gz
Documents	.pdf, .docx, .xlsx, .pptx, .csv, .txt
Web	.html, .css, .js, .json, .xml
Database	.bak, .mdf, .ndf, .ldf, .sql
Logs	.log, .evtx

12. Backup Administration

- Schedule
- Retention
- Encryption
- Storage
- Monitoring
- Integrity
- Offsite/secondary copy
- Restore testing
- RPO
- RTO

13. Skills Progression Map

CMD



PowerShell



Windows Administration



Active Directory



DNS/DHCP



File/Storage



IIS



Hyper-V



Security



Monitoring



MSSQL



Azure



Automation



DevOps

Appendix A: Automation Toolkit

The following PowerShell scripts accompany this runbook and implement the processes above directly:

Script	Purpose
Daily-HealthCheck.ps1	Runs the morning checklist across the server inventory; exports CSV + optional email summary
New-ServerProvisioning.ps1	Executes the 14-step provisioning flow and appends the result to the inventory CSV
New-UserOnboarding.ps1	Executes the HR-to-AD onboarding flow with a manager-approval gate and full audit CSV
Disk-Space-Triage.ps1	Automates the disk-full decision tree; dry-run by default, -Execute to apply cleanup
ServerInventory-Template.csv	Master server inventory schema used by the scripts above

All scripts are idempotent where possible, log every step with pass/fail status, and default to dry-run or require an explicit confirmation switch before making destructive changes.